



TJ-MU-25

Pump and mixing unit with thermostatic head
for underfloor heating manifolds · 210 mm · 20–60 °C



termojet.com.ua

WARNING! Read this manual before putting the product into service. For safety reasons, only persons familiar with its contents may install or service the product.

Application

The Termojet TJ-MU-25 pump and mixing unit connects an underfloor heating circuit to a boiler or a heat pump. The unit forms a separate low-temperature circuit: a mixing valve driven by a thermostatic head blends cooled return water into the hot flow and holds the set temperature in the floor loops.

Unlike passive units with a built-in thermostatic valve, here the temperature is measured by a **remote sensor** immersed directly in the flow. The head therefore responds to the actual circuit temperature, and the setting range is wider — **20–60 °C**.

Scope of supply

- mixing valve with an M30×1.5 thermostatic head connection;
- thermostatic head with a remote sensor and capillary;
- bypass with an adjustable setting of 0–5;
- automatic air vent G1/2";
- thermometer with a 0–80 °C scale;
- G1" connections to the manifold flow and return bars;
- mounting bracket.

The circulating pump is not supplied with the unit — it is ordered separately for a 130 mm port-to-port length, see p. 3.



The unit without the pump: capillary with the remote sensor, thermostatic head, air vent and bracket

How it works

The sensor continuously measures the flow temperature and transmits it to the head through the capillary. The head moves the valve spindle, changing the ratio of hot flow to cooled return until the temperature matches the value set on the scale.

The bypass sets how much of the flow passes the mixing point. The further it is opened, the higher the flow capacity of the unit and the heat output of the circuit.

Technical data

Flow temperature setting range	20–60 °C
Maximum primary circuit temperature	90 °C
Maximum working pressure	10 bar
Flow coefficient of the mixing valve	Kv 3
Flow coefficient with the bypass open	Kv max 4.8
Thermometer scale	0–80 °C
Port-to-port spacing of the unit	210 mm
Connection to the manifold	G1"
Pump connection	G1½", 130 mm port-to-port
Air vent connection	G1/2"
Thermostatic head connection	M30×1.5
Control method	thermostatic head with a remote sensor

Heat output and the bypass

Bypass setting	Heat output of the circuit
0 — bypass closed	10 kW
5 — bypass fully open	12.5 kW

Selecting the circulating pump

The pump is not supplied with the unit — it is ordered separately. The unit takes a pump with a G1½" connection and a port-to-port length of 130 mm. Recommended models:

Part no.	Pump
30254013	Termojet AUTO APE 25/40/130
30256013	Termojet AUTO APE 25/60/130
30258013	Termojet AUTO APE 25/80/130

Dimensions

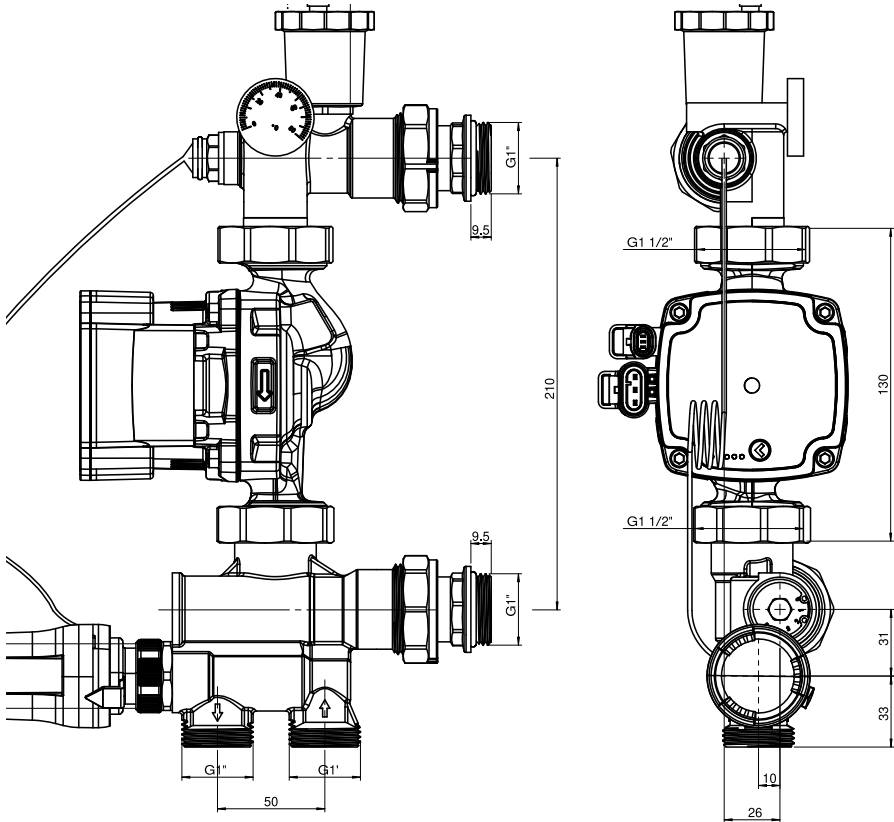


Fig. 1. Front and side views, mm. The pump is shown for reference only

Port-to-port spacing of the unit is 210 mm, the pump length is 130 mm. Connections to the manifold bars are G1" with union nuts.

Temperature setting

The flow temperature is set by turning the thermostatic head to the required mark on its scale, within 20–60 °C. Control is continuous: intermediate positions give intermediate temperatures.

The remote sensor must be fully inserted into the pocket on the flow line and secured — otherwise the head measures air temperature instead of the water temperature and the circuit overheats.

After commissioning, compare the actual flow temperature against the thermometer of the unit and correct the head setting if necessary. For a concrete screed the recommended flow temperature is no higher than 45 °C.

Setting the bypass

The greater the heat output required from the circuit, the further the bypass is opened: at setting 0 the unit delivers 10 kW, at setting 5 it delivers 12.5 kW, and intermediate settings give intermediate values. An excessively open bypass reduces the temperature difference between flow and return and makes the loops heat up less evenly.

Installation notes

- Installation is carried out by qualified trades.
- Before fitting, clean the pipework of scale and rust; a strainer at the system inlet is required for trouble-free operation.
- The direction of flow must match the arrows on the body.
- Fit the pump so that its shaft is horizontal.
- The air vent must be at the highest point of the unit and must not be shut off.
- The sensor capillary must not be kinked or stretched.
- After installation, test the system for leaks.
- Check the operation of the unit at least once every 12 months.

Using pipe wrenches during installation is not permitted.

Product data

Product	Termojet TJ-MU-25 pump and mixing unit
Manufacturer	Termojet
Distributor in Ukraine	Sofiivka-Montazh LLC, EDRPOU 37074476 EDRPOU — Ukrainian company registry code
Address	3 Kyivska St., Sofiivska Borshchahivka, Bucha district, Kyiv region, 08131, Ukraine
Contacts	+380 44 407 24 10 · termojet@sofiivka.kiev.ua termojet.com.ua

Storage and transport

Store the product in the manufacturer's packaging, in heated or unheated warehouses — no closer than one metre to any heating appliance — or under a canopy, in conditions that rule out mechanical damage.

Permissible storage and transport temperature is –20 to +50 °C. The product may be carried by any means of transport in accordance with the carriage rules in force for that mode of transport. During carriage, protect the product from impact, mechanical load and surface scratching.

Disposal

The product contains no substances hazardous to human life and health or to the environment. At the end of its service life, dispose of the product as household waste in accordance with local waste-management rules. Metal parts are suitable for remelting.

Acceptance and testing

The products covered by these instructions have been manufactured, tested and accepted in accordance with the manufacturer's current technical documentation.

Warranty

The manufacturer warrants that the product complies with safety requirements, provided that the user observes the rules set out in these instructions.

The service life of the product, provided that these instructions are observed and the required servicing is carried out, is **10 years** from the date the product is handed over to the user.

The warranty period is **24 months** from the date of sale, but no longer than the service life of the product.

The warranty covers all defects arising through the fault of the manufacturer and **does not cover** defects caused by:

- failure to observe the storage, installation, testing, operation or servicing conditions described here;
- improper transport or loading and unloading;
- exposure to substances aggressive to the materials of the product;
- fire, natural disaster or force majeure;
- acts of the user or third-party interference with the design of the product.

Products that fail within the warranty period because of a manufacturing defect are repaired or replaced free of charge. The cost of removing and transporting the faulty product is not reimbursed to the buyer. If a claim proves to be unfounded, the buyer pays for the diagnosis and examination.

To process a claim the following are required: a free-form statement giving contact details, the installation address and a description of the defect; proof of purchase; photographs of the faulty product, including photographs taken at the place of installation; a copy of the completed warranty card.

Claims are accepted at: 3 Kyivska St., Sofiivska Borshchahivka, Bucha district, Kyiv region, 08131, Ukraine. Phone +380 44 407 24 10, e-mail termojet@sofiievka.kiev.ua.

The manufacturer reserves the right to introduce changes to the design of the «Termojet TJ-MU-25 pump and mixing unit» product that do not impair its quality.

Warranty card

to delivery note No. _____ dated _____ 20____

Product: Termojet TJ-MU-25 pump and mixing unit

No.	Part number	Quantity	Notes
1			
2			
3			
4			

Warranty period – 24 months from the date of sale.

I have read and accept the warranty terms and the installation and operating instructions:

Buyer (signature)

Seller (signature)

Date of sale

Stamp or seal
of the selling organisation